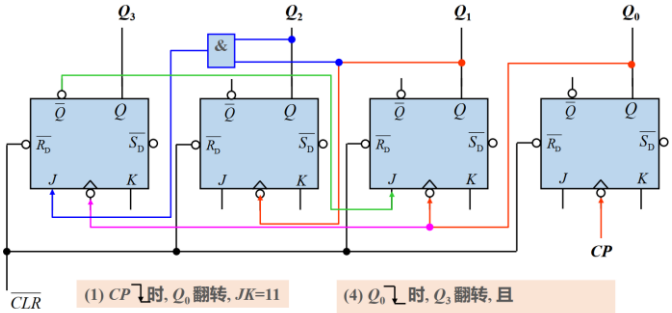


24 进制（上）　　异步十进制加法计数器（下）



原始 RS　　门控 RS　　门控 D 锁存器

$\overline{R_D}$	$\overline{S_D}$	Q
0	1	0
1	0	1
1	1	(保
0	0	1

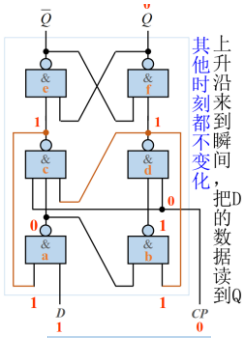
G	R	S	Q_{n+1}	说明
1	0	0	Q_n	保持
1	0	1	1	置1
1	1	0	0	清0
1	1	1	*不定	避免
0	×	×	Q_n	保持

G	D	Q
1	0	0
1	1	1
0	×	保持

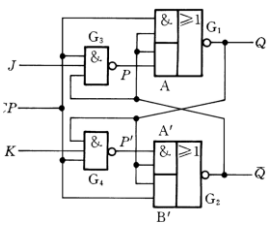
主从 SR　　主从 JK　　上升沿 D 触发器

CLK	S	R	Q	Q^{n+1}
X	X	X	X	Q^n
⊓	0	0	0	0
⊓	0	0	1	1
⊓	1	0	0	1
⊓	1	0	1	1
⊓	0	1	0	0
⊓	0	1	1	0
⊓	1	1	0	1*
⊓	1	1	1	1*

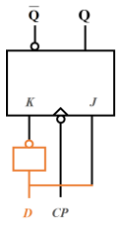
CLK	J	K	Q	Q^{n+1}
X	X	X	X	Q^n
⊓	0	0	0	0
⊓	0	0	1	1
⊓	1	0	0	1
⊓	1	0	1	1
⊓	0	1	0	0
⊓	0	1	1	0
⊓	1	1	0	1
⊓	1	1	1	0

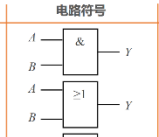


JK 触发器原理和功能表　　JK 状态方程：$Q^{n+1}=J\overline{Q^n}+\overline{K}Q^n$



J	K	CP	Q_{n+1}	说明
0	0	↑	Q_n	保持
0	1	↑	0	清0
1	0	↑	1	置1
1	1	↑	$\overline{Q_n}$	翻转
×	×	0,1,↓	Q_n	保持



门电路	电路符号	表达式
与门		$Y=AB$
或门		$Y=A+B$
非门		$Y=\overline{A}$
与非门		$Y=\overline{AB}$
或非门		$Y=\overline{A+B}$
异或门		$Y=A\oplus B$

$A(B+C)=AB+AC$

$A+BC=(A+B)(A+C)$

$A+AB=A$

$A+\overline{A}B=A+B$

$\overline{A}+AB=\overline{A}+B$

$AB+A\overline{B}=A$

$AB+\overline{A}C+BC=AB+\overline{A}C$

$\overline{ABC}=\overline{A}+\overline{B}+\overline{C}$

$\overline{A+B+C}=\overline{A}\cdot\overline{B}\cdot\overline{C}$

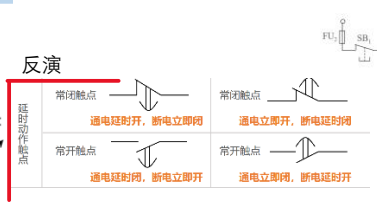
反演

$\overline{A\cdot B}=\overline{A}+\overline{B}$

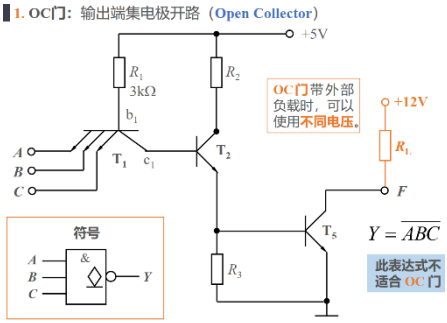
$\overline{A+B}=\overline{A}\cdot\overline{B}$

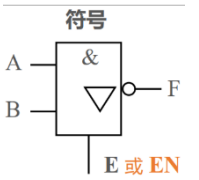
$Y=\overline{\overline{D}+\overline{A}C+\overline{B}C}=D\cdot\overline{\overline{A}C}\cdot\overline{\overline{B}C}$

化为与非式的方法



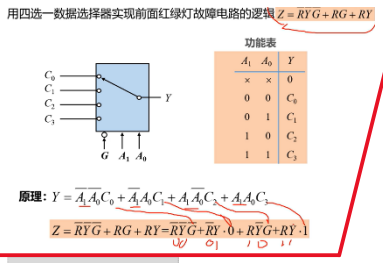
1. OC门：输出端集电极开路（Open Collector）





功能表	
$E=1$	$F=\overline{AB}$
$E=0$	输出高阻

OC 门【外接上拉源 成为与非门】 三态门的 E 模式符号与表（E 杠相反即可）



$$V_A\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)-V_B\left(\frac{1}{R_3}\right)=\frac{U_1}{R_1}+\frac{U_2}{R_2}$$

叠加原理：恒压源短，恒流源断

$$r_A=\frac{R_{AB}R_{CA}}{R_{AB}+R_{BC}+R_{CA}}\qquad r_B=\frac{R_{BC}R_{AB}}{R_{AB}+R_{BC}+R_{CA}}\qquad r_C=\frac{R_{CA}R_{BC}}{R_{AB}+R_{BC}+R_{CA}}$$

$$R_{AB}=r_A+r_B+\frac{r_Ar_B}{r_C}\qquad R_{BC}=r_B+r_C+\frac{r_Br_C}{r_A}\qquad R_{CA}=r_C+r_A+\frac{r_Cr_A}{r_B}$$

$$u=-e=L\frac{\mathrm{d}i}{\mathrm{d}t}\qquad Q=UI=I^2X_L=U^2/X_L\qquad i=C\frac{\mathrm{d}u}{\mathrm{d}t}\qquad Q=-UI$$

$$C=\frac{P}{\omega U^2}(\tan\varphi_L-\tan\varphi)$$

提高功率因数，并联电容

$$Q=\frac{U_L}{U}=\frac{U_C}{U}=\frac{X_L}{R}=\frac{X_C}{R}\qquad \mathrm{BW}=f_{C2}-f_{C1}$$

通频带（带宽）$\mathrm{BW}=\frac{R}{2\pi L}=\frac{f_0}{Q}$ 串联谐振

$$\omega_0=\sqrt{\frac{1}{LC}-\frac{R^2}{L^2}}\qquad Z=\frac{RX_C^2}{R^2+(X_L^2+X_C^2)}-\mathrm{j}\frac{R^2X_C+X_LX_C(X_L-X_C)}{R^2+(X_L^2+X_C^2)}$$

并联谐振

$$T(j\omega)=\frac{\dot{U}_o}{\dot{U}_i}=|T(j\omega)|\angle\varphi(\omega)$$

网络传递函数

三相对称电路 LP 关系与功率

$$U_L=\sqrt{3}U_P\qquad I_L=I_P\qquad P=3\,U_PI_P\cos\varphi\qquad P=\sqrt{3}\,U_L\,I_L\cos\varphi$$

$$U_L=U_P\qquad I_L=\sqrt{3}I_P$$

三要素法：

$$u_C=U+(U_0-U)\mathrm{e}^{-\frac{t}{\tau}}=u_C(\infty)+[u_C(0_+)-u_C(\infty)]\mathrm{e}^{-\frac{t}{\tau}}$$

稳态值　初始值　稳态值　时间常数

$$n_1=\frac{60f}{p}\ (\text{转/分})$$

旋转磁场转速与极对数关系

高频方波激励下的稳定 RC

$$U_1=\frac{U\mathrm{e}^{-T/\tau}}{1+\mathrm{e}^{-T/\tau}}\qquad U_2=\frac{U}{1+\mathrm{e}^{-T/\tau}}$$

脉冲分压器

$$u_{C2}(t)=\frac{R_2}{R_1+R_2}U+\left[\frac{C_1}{C_1+C_2}-\frac{R_2}{R_1+R_2}\right]U\mathrm{e}^{-t/\tau}$$

电机 Y 起动的电流和力矩都降到三角的 1/3

$$s=\frac{n_1-n}{n_1}\qquad f_2=sf_1$$

转差率、感生电流频率

$$I_2=\frac{sE_{20}}{\sqrt{R_2^2+(sX_{20})^2}}$$

转子电流；启动时 s=1

$$T_{st}=K\frac{R_2}{R_2^2+(X_{20})^2}\cdot U_1^2$$

起动转矩

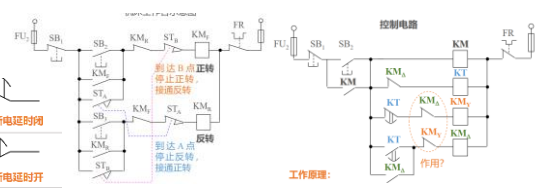
$$T=K\frac{sR_2}{R_2^2+(sX_{20})^2}\cdot U_1^2$$

额定（下）

最大（下）

起动能力：T_{st}/T_N

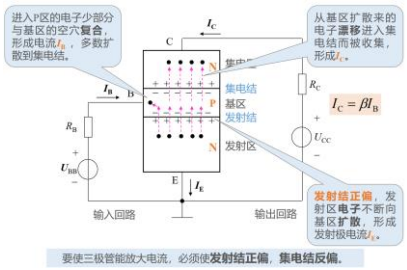
$$T_N=\frac{P_N}{\frac{2\pi n_N}{60}}=9550\frac{P_N\,(\text{千瓦})}{n_N\,(\text{转/分})}\ (\mathrm{N\cdot m})\qquad s_m=\frac{R_2}{X_{20}}\qquad T_m=KU_1^2\frac{1}{2X_{20}}$$



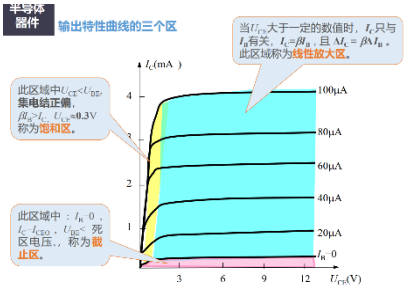
导通压降：硅管0.6~0.7V,锗管0.2~0.3V。 死区电压 硅管0.5V,锗管0.2V。

$$i = I_S(e^{\frac{u}{U_T}} - 1)$$

无负反馈的2*共射极。



要使三极管能放大电流，必须使**发射结正偏，集电结反偏。**



第1级电压放大倍数：

$$\dot{A}_{u1} = \frac{\dot{U}_{o1}}{\dot{U}_i} = -\beta_1 \frac{R_{C1} // r_{i2}}{r_{be1}}$$

其中 $r_{i2} = R_{21} // R_{22} // r_{be2}$

总输入电阻：

$$r_i = r_{i1} = R_{11} // R_{12} // r_{be1}$$

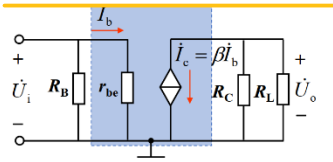
第2级电压放大倍数：

$$\dot{A}_{u2} = \frac{\dot{U}_o}{\dot{U}_{i2}} = -\beta_2 \frac{R_{C2} // R_L}{r_{be2}}$$

总输出电阻：

$$r_o = r_{o2} = R_{C2}$$

基本共射极放大电路（下式）

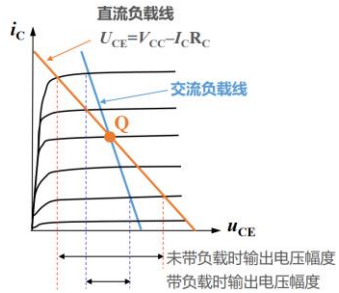


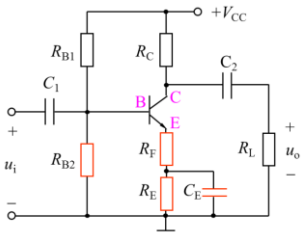
$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta R'_L}{r_{be}}$$

输入电阻：$r_i = \frac{\dot{U}_i}{\dot{I}_i} = R_B // r_{be}$　输出电阻：$r_o = R_C$

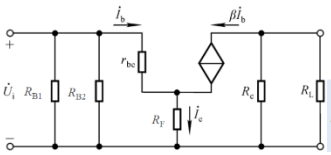
$$r_{be} = 300(\Omega) + (1+\beta)\frac{26(\text{mV})}{I_E(\text{mA})}$$

射极交流负反馈电阻放大器（右下图）





(1) Q点过低，信号进入截止区，容易发生**截止失真**。　(2) Q点过高，信号进入饱和区，容易发生**饱和失真**。

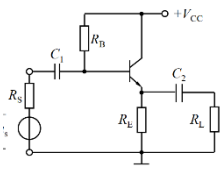


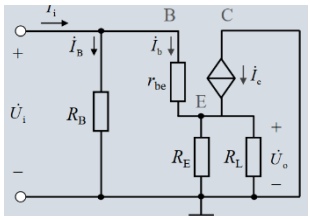
$$\dot{A}_{u载} = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta R'_L}{r_{be} + (\beta+1)R_F}$$

$$r_i = \frac{\dot{U}_i}{\dot{I}_i} = R_{B1} // R_{B2} // [r_{be} + (\beta+1)R_F]$$

Ro=Rc 不变

共集电极放大器（射随）

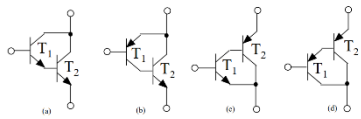




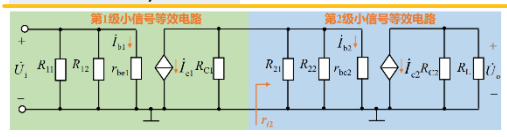
$$\dot{A}_u = \frac{(\beta+1)R'_L}{r_{be} + (\beta+1)R'_L}$$

$$r_i = R_B // [r_{be} + (\beta+1)R'_L]$$

$$r_o = R_E // \frac{r_{be} + R_S // R_B}{\beta+1}$$

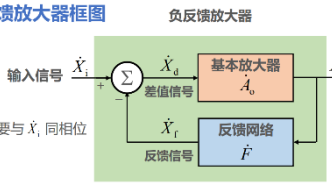


复合管，箭头朝外 NPN，类型与 T1 一致



前一级 ro 作为后一级电源内阻，后一级 ro 是前的负载，分别计算。

负反馈放大器框图



① 串联反馈使电路的输入电阻增加：$r_{if} = (1 + A_o \dot{F}) r_i$

② 并联反馈使电路的输入电阻减小：$r_{if} = \frac{r_i}{(1 + A_o \dot{F})}$

③ 电压反馈使电路的输出电阻减小：$r_{of} = \frac{r_o}{(1 + A_o \dot{F})}$

④ 电流反馈使电路的输出电阻增加

开环放大倍数 $\dot{A}_o = \frac{\dot{X}_o}{\dot{X}_d}$

反馈系数 $\dot{F} = \frac{\dot{X}_f}{\dot{X}_o}$

差值信号 $\dot{X}_d = \dot{X}_i - \dot{X}_f$

闭环放大倍数 $\dot{A}_F = \frac{\dot{X}_o}{\dot{X}_i}$

闭环放大倍数$\dot{A}_F$与$\dot{A}_o$和$\dot{F}$的关系

$$\dot{A}_F = \frac{\dot{X}_o}{\dot{X}_i} = \frac{\dot{X}_o}{\dot{X}_d + \dot{X}_f} = \frac{\dot{X}_o}{\frac{\dot{X}_o}{\dot{A}_o} + \dot{F}\dot{X}_o}$$

$$\dot{A}_F = \frac{\dot{A}_o}{1 + \dot{F}\dot{A}_o}$$

$$\dot{B}_f = (1 + A_o \dot{F}) B_o$$

将放大器（交流通道）**输出电压置0**，若反馈信号**消失**，输入和反馈在**相同端** → **并联**反馈；是**电压反馈**；若反馈信号**不消失**，则为**电流反馈**。

输入和反馈在**不同端** → **串联**反馈。

(1) C点电位与B电位变化极性相反

(2) E点电位与B电位变化极性相同

(3) 若B点电位固定，则C点电位与E点电位变化极性相同

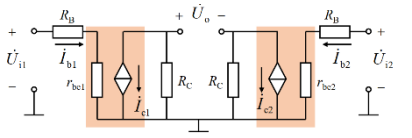
反馈组态	$\dot{A}_{uf}$ 或 $\dot{A}_{usf}$
电压串联	$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{\dot{U}_o}{\dot{U}_f} = \frac{1}{\dot{F}_{uu}}$
电压并联	$\dot{A}_{usf} = \frac{\dot{U}_o}{\dot{U}_s} \approx \frac{1}{\dot{F}_{iu}} \cdot \frac{1}{R_s}$
电流串联	$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{1}{\dot{F}_{ui}} \cdot R'_L$
电流并联	$\dot{A}_{usf} = \frac{\dot{U}_o}{\dot{U}_s} \approx \frac{1}{\dot{F}_{ii}} \cdot \frac{R'_L}{R_s}$

$$A_c = \frac{u_{oc}}{u_{i1}} \qquad A_d = \frac{u_o}{u_{i1} - u_{i2}} = \frac{u_o}{2u_{i1}}$$

$$K_{CMRR} = 20\log\left|\frac{A_d}{A_c}\right| \text{ dB}$$

$$A_d = -\frac{\beta\left(R_C // \frac{R_L}{2}\right)}{R_B + r_{be}}$$

长尾式带载差模放大



双端输出

$$A_d = -\frac{\beta(R_C // (R_L/2))}{R_B + r_{be}}$$

单端输出

$$A_d = -\frac{\beta(R_C // R_L)}{2(R_B + r_{be})}$$

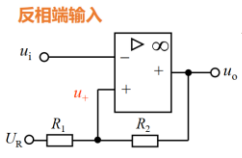
$$A_d = \frac{\dot{U}_o}{2\dot{U}_{i1}} = -\frac{\beta R_C}{R_B + r_{be}}$$

相当于一边的放大倍数

当R_B=0时，$A_d = -\frac{\beta R_C}{r_{be}}$

理想运放输入端的虚短与虚断 i=0，u 相等

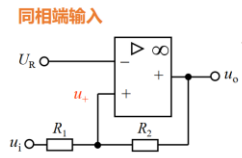
反相端输入



$$U_{TH} = \frac{R_1}{R_1+R_2}U_{OM} + \frac{R_2}{R_1+R_2}U_R$$

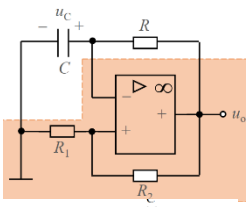
$$U_{TL} = -\frac{R_1}{R_1+R_2}U_{OM} + \frac{R_2}{R_1+R_2}U_R$$

同相端输入



$$U_{TH} = \frac{R_1}{R_2}U_{OM} + \frac{R_1+R_2}{R_2}U_R$$

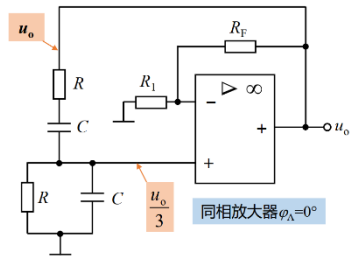
$$U_{TL} = -\frac{R_1}{R_2}U_{OM} + \frac{R_1+R_2}{R_2}U_R$$



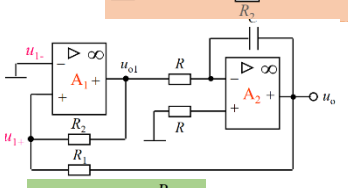
$$T = 2RC\ln\left(1 + \frac{2R_1}{R_2}\right)$$

方波发生器

uc 充电时 o 高电平



文氏 1/3 RF= 2R1



$$T = 2(t_3 - t_2) = 4\frac{R_1}{R_2}RC$$

三角波

$$f_0 = \frac{1}{2\pi RC} \qquad f_0 \approx \frac{1}{2\pi\sqrt{LC}}$$

三点式 LC 电路：

若中间点交流接地，则首端与尾端相位相反。若首端或尾端交流接地，则其他两端相位相同。频率条件见右上。